

실습 가이드: OpenFOAM cavity flow

2026 한국연소학회 가을 Tutorial
Next-Gen Combustion

Cavity Flow

• Tutorial 1: cavity flow

- Describe how to pre-process, run and post-process a case involving isothermal, incompressible flow
- **Lid-driven cavity flow** (see Fig. 1)
 - 2D square domain
 - Top wall moves in the x-direction at 1 m/s and other 3 walls are stationary.
 - incompressibleFluid modular solver:
for laminar, isothermal, incompressible flow.
 - General procedure for CFD simulations.
 - Pre-processing (e.g., mesh generation)
 - Run the simulation
 - Post-processing

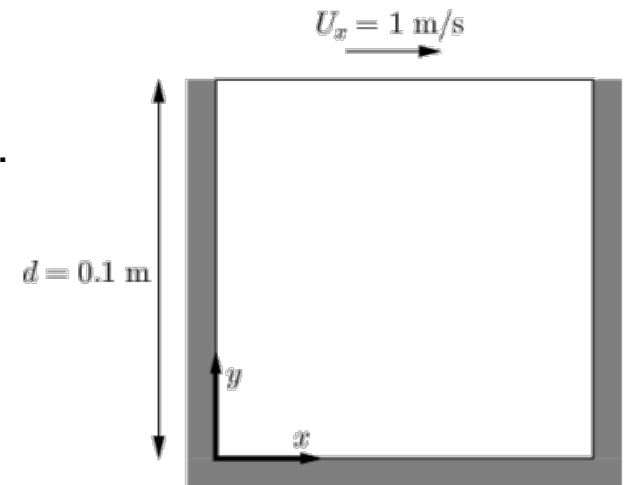


Figure 1. Geometry of the lid driven cavity

Cavity Flow

- **Governing equation of incompressibleFluid**

- A solver module for transient, incompressible, isothermal laminar flow simulation
- For incompressible flow, the pressure is expressed as kinematic pressure $\frac{p}{\rho}$
- Governing equations:

rhoEqn: $\nabla \cdot \mathbf{u} = 0$

UEqn: $\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) - \nabla \cdot (\nu \nabla \mathbf{u}) = -\nabla \frac{p}{\rho}$

Cavity Flow

- Pre-processing

- Get the tutorial files
- The recommended location is your OpenFOAM run directory

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Command Line

```
mkdir -p $FOAM_RUN
```

```
cd $FOAM_RUN
```

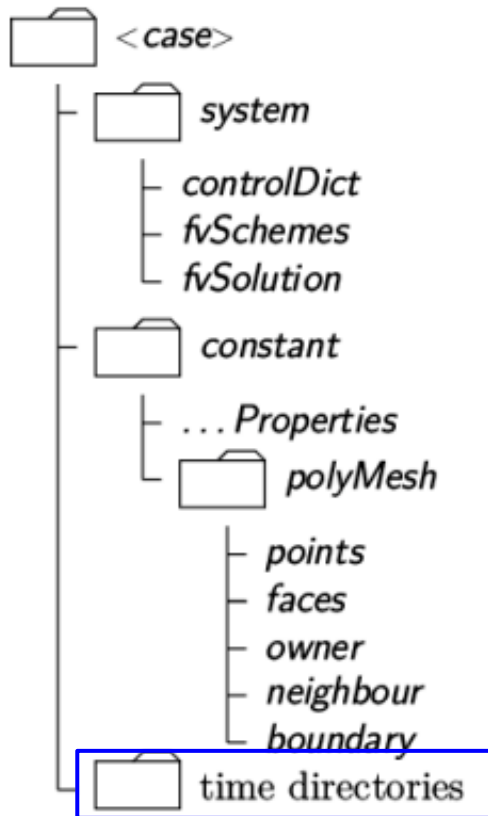
```
git clone https://github.com/CCERL/KOSCO-2026-NextGen-Combustion.git
```

```
cd KOSCO-2026-NextGen-Combustion/Tutorial/1.Cavity
```

```
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run$ git clone https://github.com/CCERL/KOSCO-2026-NextGen-Combustion.git
Cloning into 'KOSCO-2026-NextGen-Combustion'...
remote: Enumerating objects: 506, done.
remote: Counting objects: 100% (208/208), done.
remote: Compressing objects: 100% (177/177), done.
remote: Total 506 (delta 83), reused 81 (delta 31), pack-reused 298 (from 1)
Receiving objects: 100% (506/506), 4.46 MiB | 13.32 MiB/s, done.
Resolving deltas: 100% (226/226), done.
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run$ cd KOSCO-2026-NextGen-Combustion/Tutorial/1.Cavity
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/1.Cavity$ ll
total 28
drwxr-xr-x 5 ghkrk ghkrk 4096 Oct  6 10:56 ./
drwxr-xr-x 6 ghkrk ghkrk 4096 Oct  6 10:56 ../
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 0/
-rwxr-xr-x 1 ghkrk ghkrk 257 Oct  6 10:56 Allclean*
-rwxr-xr-x 1 ghkrk ghkrk 326 Oct  6 10:56 Allrun*
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 constant/
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 system/
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/1.Cavity$
```

Cavity Flow

• File structure of OpenFOAM case



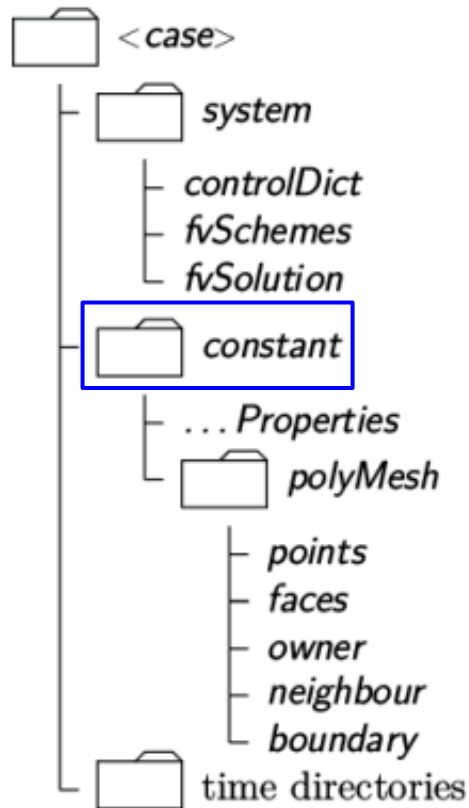
- time directory: contains individual files of data for particular fields (e.g., U and p). Each data files can be either
 - **Initial fields** and **boundary conditions** that the user must specify to define the problem
 - Output file
- The name of each time directory is based on the simulated time at which the data is written.

```
4096 Sep 30 19:18 0
4096 Sep 30 19:18 constant
4096 Sep 30 19:18 system
```

Overview of file structure of OpenFOAM case*

Cavity Flow

• File structure of OpenFOAM case

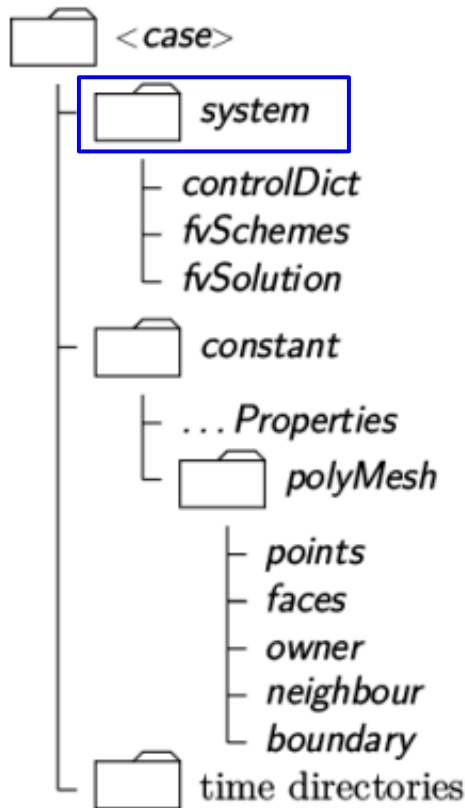


- constant directory: contains
 - polyMesh : Full collection of the mesh files
 - Physical properties: e.g., viscosity and thermophysical properties (e.g.,
 - Physical models: e.g., laminar/turbulence, combustion, and chemical reaction models
 - Example dictionary file: momentumTransport, physicalProperties, chemistryProperties

Overview of file structure of OpenFOAM case*

Cavity Flow

• File structure of OpenFOAM cases



- system directory: contains **at least** the following 3 files:
 - `controlDict`: run control parameters are set including start/end time, time step and parameters for data output.
 - `fvSchemes`: discretization schemes used in the solution
 - `fvSolution`: the equation solvers, tolerances and other algorithm controls are set for the run.

Overview of file structure of OpenFOAM case*

Cavity Flow

● Preprocessing: Boundary and Initial Conditions

- Every field file consists of *internalField* and *boundaryField*.

```
/*----- C++ -----*/
//
//  F i e l d
//  O p e r a t i o n
//  A n d
//  M a n i p u l a t i o n
//
//  OpenFOAM: The Open Source CFD Toolbox
//  Website:  https://openfoam.org
//  Version:  12
//
FoamFile
{
    format      ascii;
    class       volVectorField;
    object      U;
}
// ***** //

dimensions      [0 1 -1 0 0 0 0];

internalField    uniform (0 0 0);

boundaryField
{
    movingWall
    {
        type      fixedValue;
        value      uniform (1 0 0);
    }

    fixedWalls
    {
        type      noSlip;
    }

    frontAndBack
    {
        type      empty;
    }
}
// ***** //
```

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Command Line

vi 0/U

To exit vi editor, type **:q**

:wq	write and quit the file
:w	write
:q	quit the file
:q!	quit the file (ignore the change)

Physical dimension of the field.

Addition, subtraction and equality are only physically meaningful for properties of the same dimensional units. (As a safeguard against implementing a meaningless operation)

[kg m s K mol A cd].

Cavity Flow

● Preprocessing: Boundary and Initial Conditions

```
/*----- C++ -----*/
//
//  F i e l d
//  O p e r a t i o n
//  A n d
//  M a n i p u l a t i o n
//
// OpenFOAM: The Open Source CFD Toolbox
// Website: https://openfoam.org
// Version: 12
//
FoamFile
{
    format      ascii;
    class       volVectorField;
    object      U;
}
// ***** //

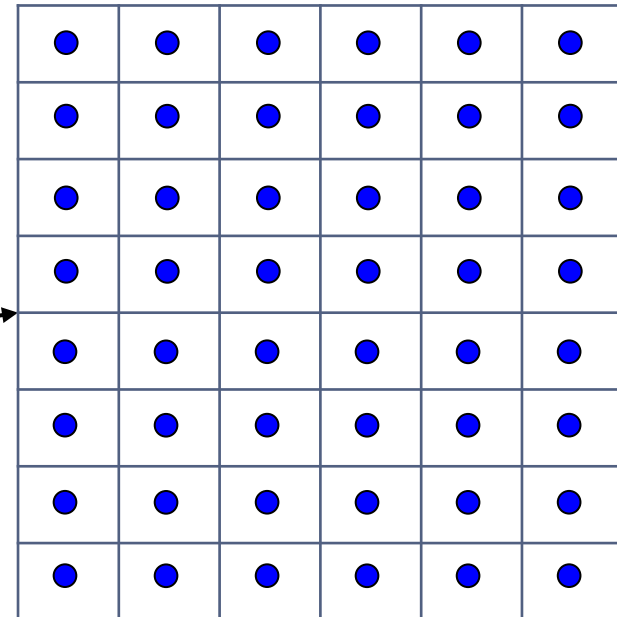
dimensions      [0 1 -1 0 0 0];

internalField   uniform (0 0 0);

boundaryField
{
    movingWall
    {
        type      fixedValue;
        value      uniform (1 0 0);
    }

    fixedWalls
    {
        type      noSlip;
    }

    frontAndBack
    {
        type      empty;
    }
}
// ***** //
```



internalField of U is uniformly set to (0 0 0) m/s
- Initial condition

Cavity Flow

● Preprocessing: Boundary and Initial Conditions

```
/*----- C++ -----*/
//
//  F i e l d
//  O p e r a t i o n
//  A n d
//  M a n i p u l a t i o n
//
//  OpenFOAM: The Open Source CFD Toolbox
//  Website: https://openfoam.org
//  Version: 12
//
FoamFile
{
    format      ascii;
    class       volVectorField;
    object      U;
}
// ***** //

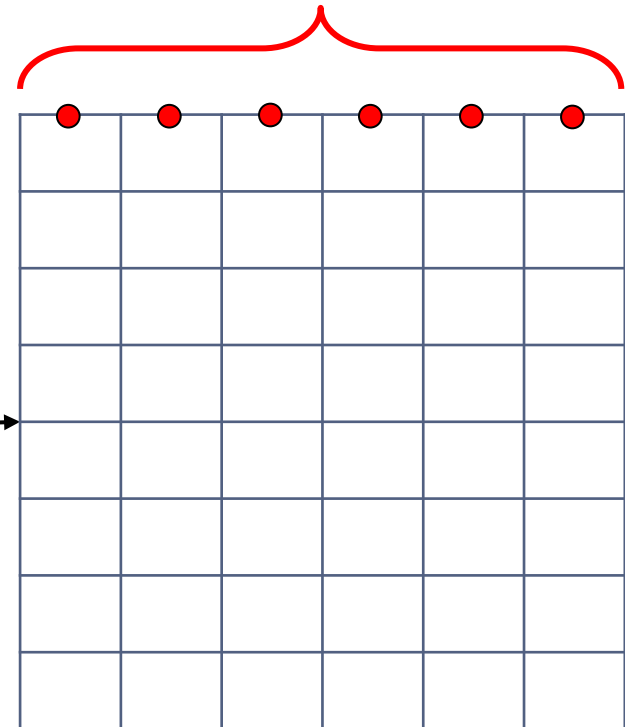
dimensions      [0 1 -1 0 0 0];
internalField    uniform (0 0 0);

boundaryField
{
    movingWall
    {
        type      fixedValue;
        value      uniform (1 0 0);
    }

    fixedWalls
    {
        type      noSlip;
    }

    frontAndBack
    {
        type      empty;
    }
}
// ***** //
```

U at the movingWall
(1 0 0) m/s



Cavity Flow

● Preprocessing: Boundary and Initial Conditions

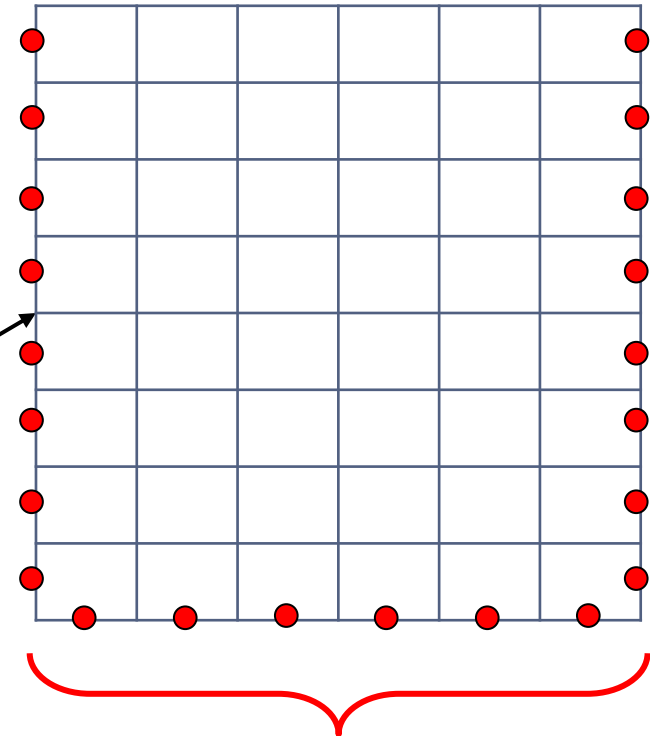
```
/*----- C++ -----*/
//
//  F i e l d
//  O p e r a t i o n
//  A n d
//  M a n i p u l a t i o n
//
//  OpenFOAM: The Open Source CFD Toolbox
//  Website: https://openfoam.org
//  Version: 12
//
FoamFile
{
    format      ascii;
    class       volVectorField;
    object      U;
}
// ***** //

dimensions      [0 1 -1 0 0 0];
internalField   uniform (0 0 0);

boundaryField
{
    movingWall
    {
        type      fixedValue;
        value      uniform (1 0 0);
    }

    fixedWalls
    {
        type      noSlip;
    }

    frontAndBack
    {
        type      empty;
    }
}
// ***** //
```



U at the remaining wall
no slip boundary condition

Cavity Flow

● Preprocessing: Boundary and Initial Conditions

```
/*----- C++ -----*/
=====
\\      /   F i e l d           OpenFOAM: The Open Source CFD Toolbox
 \\    /     O p e r a t i o n   Website: https://openfoam.org
  \\  /      A n d               Version: 12
   \\//       M a n i p u l a t i o n
=====
FoamFile
{
    format            ascii;
    class             volVectorField;
    object            U;
}
// *****

dimensions          [0 1 -1 0 0 0];

internalField        uniform (0 0 0);

boundaryField
{
    movingWall
    {
        type          fixedValue;
        value          uniform (1 0 0);
    }

    fixedWalls
    {
        type           noSlip;
    }

    frontAndBack
    {
        type            empty;
    }
}

// *****
```

frontAndBack, which are the faces normal to z , is set to empty

Cavity Flow

● Preprocessing: Boundary and Initial Conditions

```
/*----- C++ -----*/
//
// \V\  F ield      | OpenFOAM: The Open Source CFD Toolbox
//  \O\  O peration | Website: https://openfoam.org
//   \A\  A nd      | Version: 12
//    \M\  M anipulation
//
FoamFile
{
    format      ascii;
    class       volScalarField;
    object      p;
}
// *****

dimensions      [0 2 -2 0 0 0];

internalField   uniform 0;

boundaryField
{
    movingWall
    {
        type      zeroGradient;
    }

    fixedWalls
    {
        type      zeroGradient;
    }

    frontAndBack
    {
        type      empty;
    }
}
// *****
```

3 Command Line

vi 0/p

kinematic pressure : $\frac{p}{\rho}$

$$\left[\frac{\text{Pa}}{\text{kg/m}^3} = \text{m}^2 \cdot \text{s}^{-2} \right]$$

Unit of kinematic pressure.

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) - \nabla \cdot (\nu \nabla \mathbf{u}) = -\nabla \frac{p}{\rho}$$

Cavity Flow

● Preprocessing: Boundary and Initial Conditions

```
/*-----* C++ -*-----*/
//
// \V\  F ield      | OpenFOAM: The Open Source CFD Toolbox
//  \O\  O peration | Website: https://openfoam.org
//   \A\  A nd      | Version: 12
//    \M\  M anipulation
//
FoamFile
{
    format      ascii;
    class       volScalarField;
    object      p;
}
// ***** //

dimensions      [0 2 -2 0 0 0];

internalField   uniform 0;

boundaryField
{
    movingWall
    {
        type      zeroGradient;
    }

    fixedWalls
    {
        type      zeroGradient;
    }

    frontAndBack
    {
        type      empty;
    }
}
// ***** //
```

In N-S equation, gradient of kinematic pressure is important such that we give a reference value 0 as a uniform internalField.

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) - \nabla \cdot (\nu \nabla \mathbf{u}) = -\nabla \frac{p}{\rho}$$

Cavity Flow

- Preprocessing: Boundary and Initial Conditions

```
/*----- C++ -----*/
//
// \=====
//  F i e l d
//  O p e r a t i o n
//  A n d
//  M a n i p u l a t i o n
//
// OpenFOAM: The Open Source CFD Toolbox
// Website: https://openfoam.org
// Version: 12
//
//-----*/
FoamFile
{
    format      ascii;
    class       volScalarField;
    object      p;
}
// ***** //

dimensions      [0 2 -2 0 0 0];

internalField   uniform 0;

boundaryField
{
    movingWall
    {
        type      zeroGradient;
    }
    fixedWalls
    {
        type      zeroGradient;
    }
    frontAndBack
    {
        type      empty;
    }
}
// ***** //
```

$$\nabla \frac{p}{\rho} = 0 \text{ at the wall}$$

Cavity Flow

- Preprocessing: Physical properties

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) - \nabla \cdot (\nu \nabla \mathbf{u}) = -\nabla \frac{p}{\rho}$$

Kinematic viscosity

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Command Line

vi constant/physicalProperties

```
/*----- C++ -----*/
//
//  F ield
//  O peration
//  A nd
//  M anipulation
//
//  OpenFOAM: The Open Source CFD Toolbox
//  Website:  https://openfoam.org
//  Version:  12
//
FoamFile
{
    format      ascii;
    class       dictionary;
    location    "constant";
    object      physicalProperties;
}
// *****

viscosityModel constant;
nu             0.01 [m^2/s];
// *****
```

$\nu \text{ [m}^2\text{/s]} = 0.01$

Cavity Flow

- **Preprocessing: Simulation control**

- In the *system* directory,
 - **controlDict**: input data relating to the control of time and reading and writing of the solution
 - fvSchemes, fvSolution: Settings for numerical discretization and solution methods

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Command Line

vi system/controlDict

```

/*----- C++ -----*/
//
// \      /
//  \    /
//   \  /
//    \/
//
Field
Operation
And
Manipulation

OpenFOAM: The Open Source CFD Toolbox
Website: https://openfoam.org
Version: 12

FoamFile
{
    format      ascii;
    class       dictionary;
    location    "system";
    object      controlDict;
}

// ***** Solver driver ***** //
application    foamRun;

// ***** Modular solver ***** //
solver         incompressibleFluid;

startFrom      startTime;

startTime      0;

stopAt         endTime;

endTime        0.1;
deltaT         1e-4;

writeControl    timeStep;

writeInterval   50;

purgeWrite     0;

writeFormat     ascii;

writePrecision  6;

writeCompression off;

timeFormat      general;

timePrecision   6;

runTimeModifiable true;

```

Cavity Flow

- **Preprocessing: Mesh Generation**

- The standard mesh generator utility: blockMesh

⑥

Command Line

blockMesh

```
Creating polyMesh from blockMesh
Creating patches
Creating cells
Creating points with scale 1
  Block 0 cell size :
    i : 0.0002 .. 0.0002
    j : 0.0005
    k : 0.02

No patch pairs to merge

Writing polyMesh
-----
Mesh Information
-----
  boundingBox: (0 -0.01 -0.01) (0.02 0.01 0.01)
  nPoints: 8282
  nCells: 4000
  nFaces: 16140
  nInternalFaces: 7860
-----
Patches
-----
  patch 0 (start: 7860 size: 40) name: fuel
  patch 1 (start: 7900 size: 40) name: air
  patch 2 (start: 7940 size: 200) name: outlet
  patch 3 (start: 8140 size: 8000) name: frontAndBack
```

End

Cavity Flow

- Solving: Run the simulation

- Run the simulation after preprocessing

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Command Line

```
foamRun
```

```
Courant Number mean: 0.00440159 max: 0.0169966  
Time = 0.0999s
```

```
smoothSolver: Solving for Ux, Initial residual = 6.32875e-06, Final residual = 2.44319e-09, No Iterations 3  
smoothSolver: Solving for Uy, Initial residual = 8.26489e-06, Final residual = 3.23829e-09, No Iterations 3  
GAMG: Solving for p, Initial residual = 1.49659e-06, Final residual = 1.21309e-07, No Iterations 2  
time step continuity errors : sum local = 4.21099e-12, global = -1.42358e-21  
GAMG: Solving for p, Initial residual = 1.2636e-07, Final residual = 5.886e-09, No Iterations 2  
time step continuity errors : sum local = 2.45565e-13, global = 1.1114e-20, cumulative = 4.25276e-19  
ExecutionTime = 0.842519 s ClockTime = 0 s
```

```
Courant Number mean: 0.00440163 max: 0.0169966  
Time = 0.1s
```

```
smoothSolver: Solving for Ux, Initial residual = 6.2959e-06, Final residual = 2.43046e-09, No Iterations 3  
smoothSolver: Solving for Uy, Initial residual = 8.22187e-06, Final residual = 3.22143e-09, No Iterations 3  
GAMG: Solving for p, Initial residual = 1.48877e-06, Final residual = 1.20682e-07, No Iterations 2  
time step continuity errors : sum local = 4.18918e-12, global = 1.96869e-22  
GAMG: Solving for p, Initial residual = 1.25698e-07, Final residual = 5.85678e-09, No Iterations 2  
time step continuity errors : sum local = 2.44337e-13, global = -7.36604e-21, cumulative = 4.1791e-19  
ExecutionTime = 0.84419 s ClockTime = 0 s
```

```
End
```

Cavity Flow

- Solving: Run the simulation

- You can see the calculated time directories

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Command Line

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```
total 112
drwxr-xr-x 25 ghkrk ghkrk 4096 Oct 7 19:35 ./
drwxr-xr-x 6 ghkrk ghkrk 4096 Oct 6 14:18 ../
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct 6 10:56 0/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 7 19:35 0.005/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 7 19:35 0.01/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 7 19:35 0.015/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 7 19:35 0.02/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 7 19:35 0.025/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 7 19:35 0.03/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 7 19:35 0.035/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 7 19:35 0.04/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 7 19:35 0.045/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 7 19:35 0.05/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 7 19:35 0.055/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 7 19:35 0.06/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 7 19:35 0.065/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 7 19:35 0.07/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 7 19:35 0.075/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 7 19:35 0.08/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 7 19:35 0.085/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 7 19:35 0.09/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 7 19:35 0.095/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 7 19:35 0.1/
-rwxr-xr-x 1 ghkrk ghkrk 257 Oct 6 10:56 Allclean*
-rwxr-xr-x 1 ghkrk ghkrk 344 Oct 7 19:31 Allrun*
-rw-r--r-- 1 ghkrk ghkrk 1602 Oct 6 14:33 README.md
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 7 19:35 constant/
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct 7 19:35 system/
```

Saved results

Cavity Flow

• Post-Processing

- make dummy file to use Paraview

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Command Line

```
touch cavity.foam
```

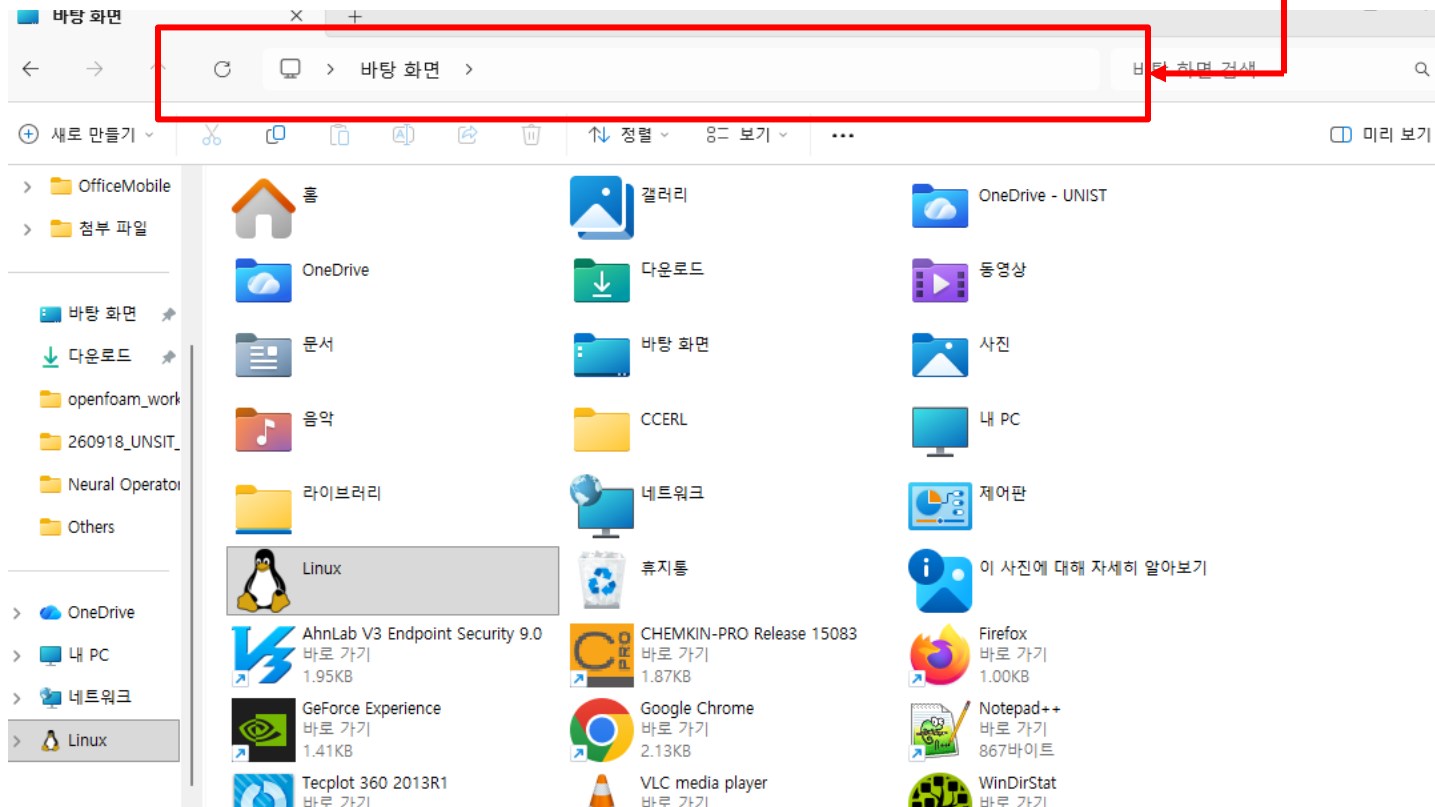
```
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/1.Cavity$ touch cavity.foam
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/1.Cavity$ ll
total 108
drwxr-xr-x 25 ghkrk ghkrk 4096 Oct  6 10:50 ./
drwxr-xr-x  6 ghkrk ghkrk 4096 Oct  5 16:31 ../
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  5 16:43 0/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 0.5/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 1/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 1.5/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 10/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 2/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 2.5/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 3/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 3.5/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 4/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 4.5/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 5/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 5.5/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 6/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 6.5/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 7/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 7.5/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 8/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 8.5/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 9/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 9.5/
-rwxr-xr-x  1 ghkrk ghkrk  257 Oct  5 16:31 Allclean*
-rwxr-xr-x  1 ghkrk ghkrk  326 Oct  5 16:31 Allmult*
-rw-r--r--  1 ghkrk ghkrk    0 Oct  6 10:50 cavity.foam
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  6 10:49 constant/
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  5 16:53 system/
```

Cavity Flow

● Post-Processing

- Find your WSL directory

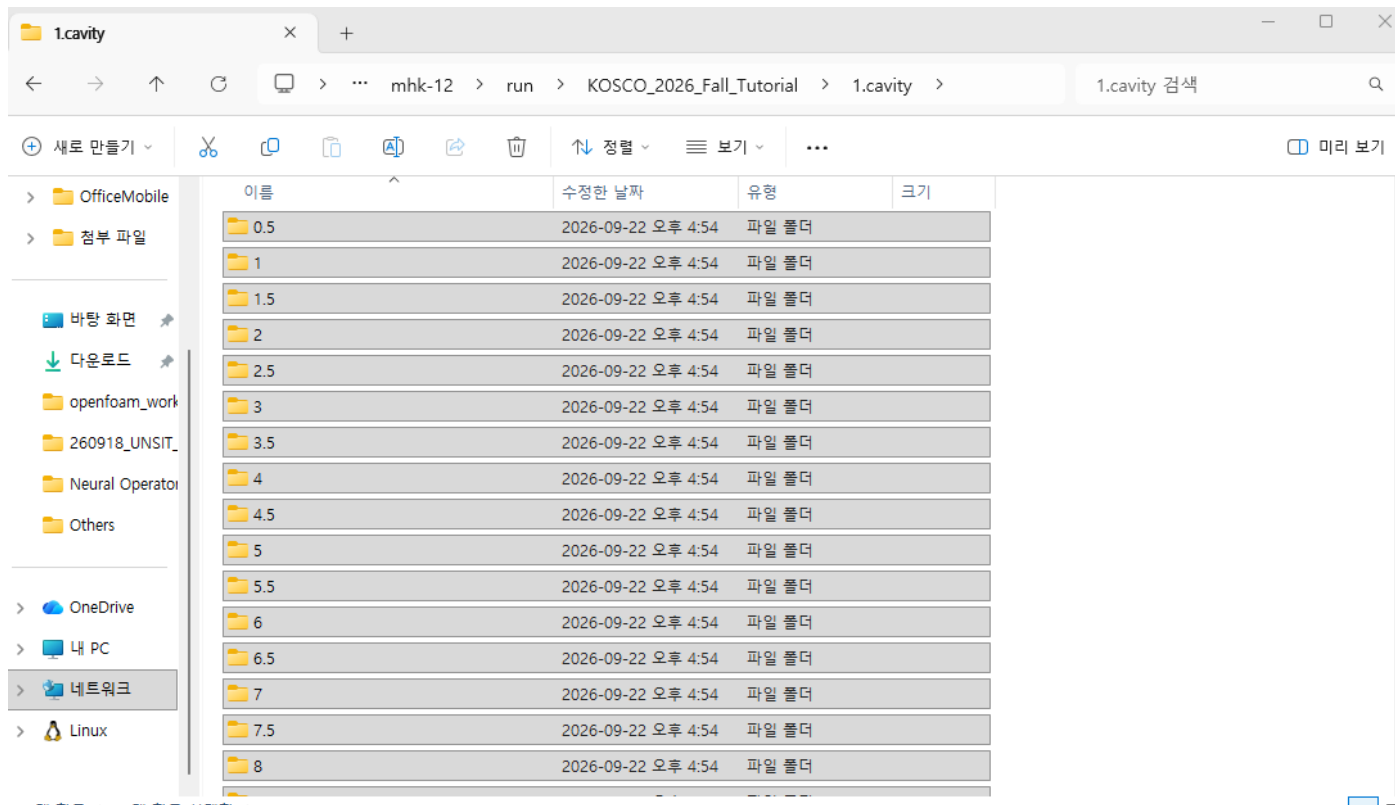
\\wsl.localhost\Ubuntu-22.04\home\Username\OpenFOAM\Username-12\run\KOSCO_2026_NextGen-Combustion\1.Cavity



Cavity Flow

● Post-Processing

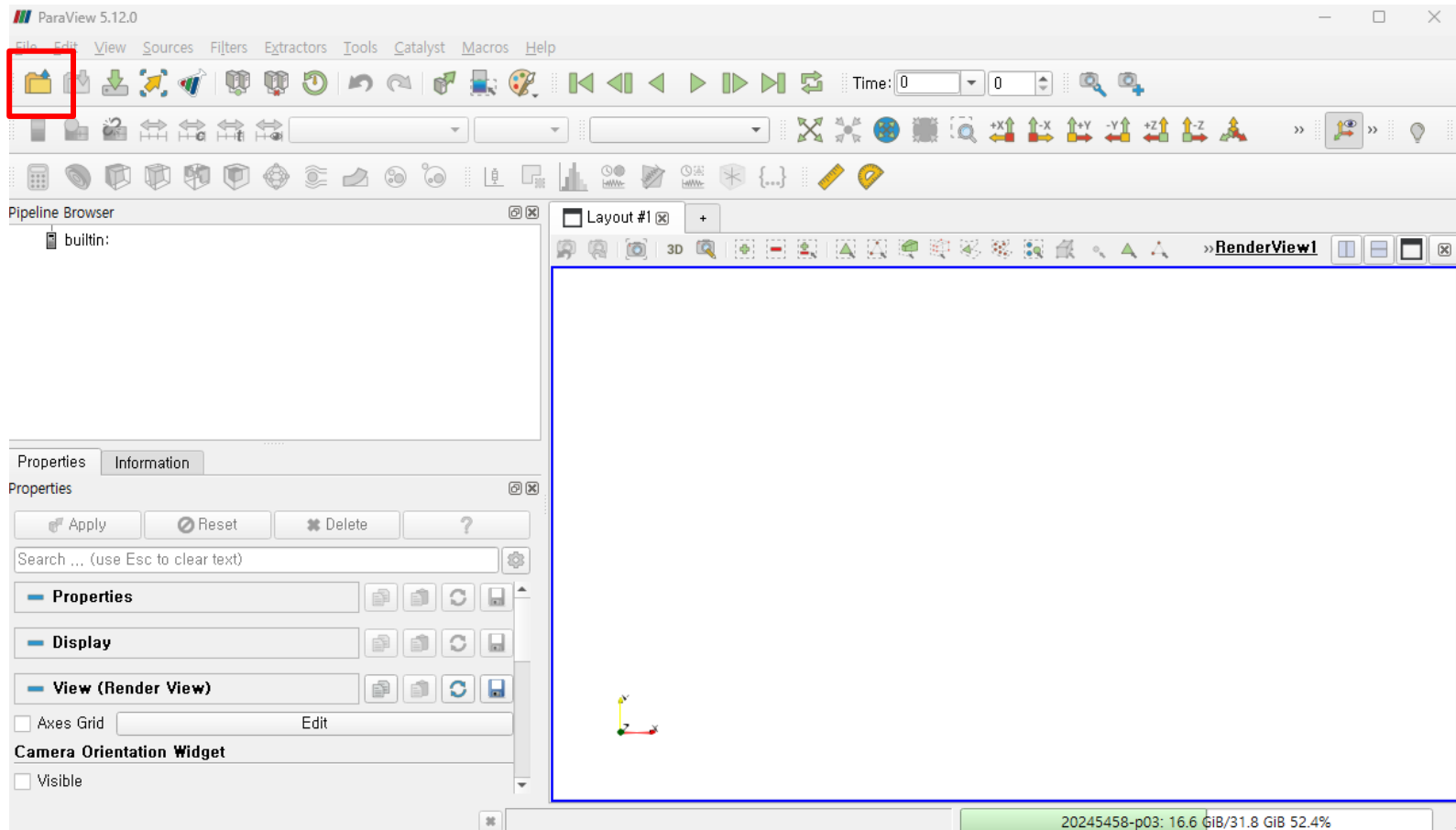
- Copy and paste the whole files anywhere under C: drive or D: drive



Cavity Flow

- **Post-Processing**

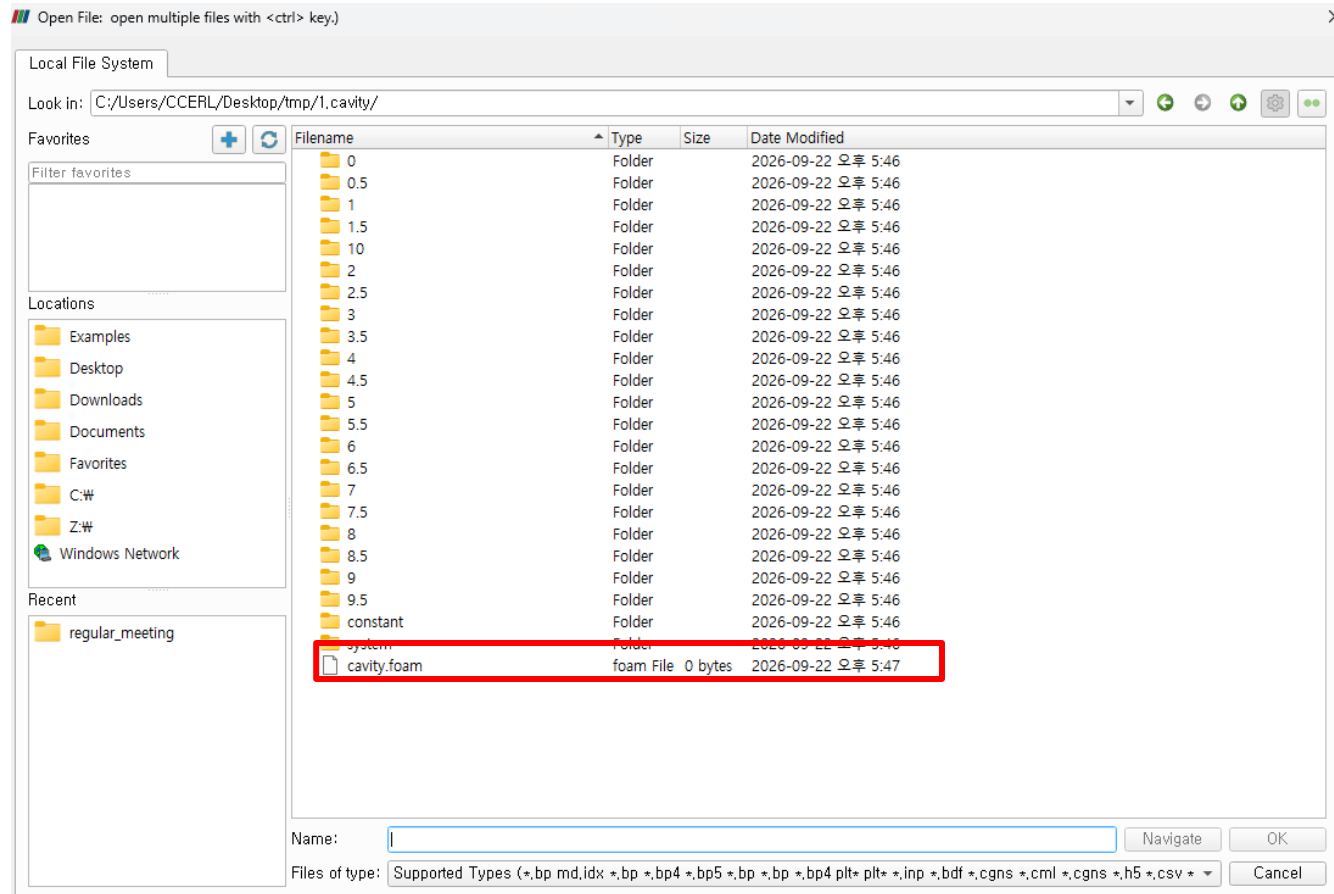
- Open or Browse file



Cavity Flow

- Post-Processing

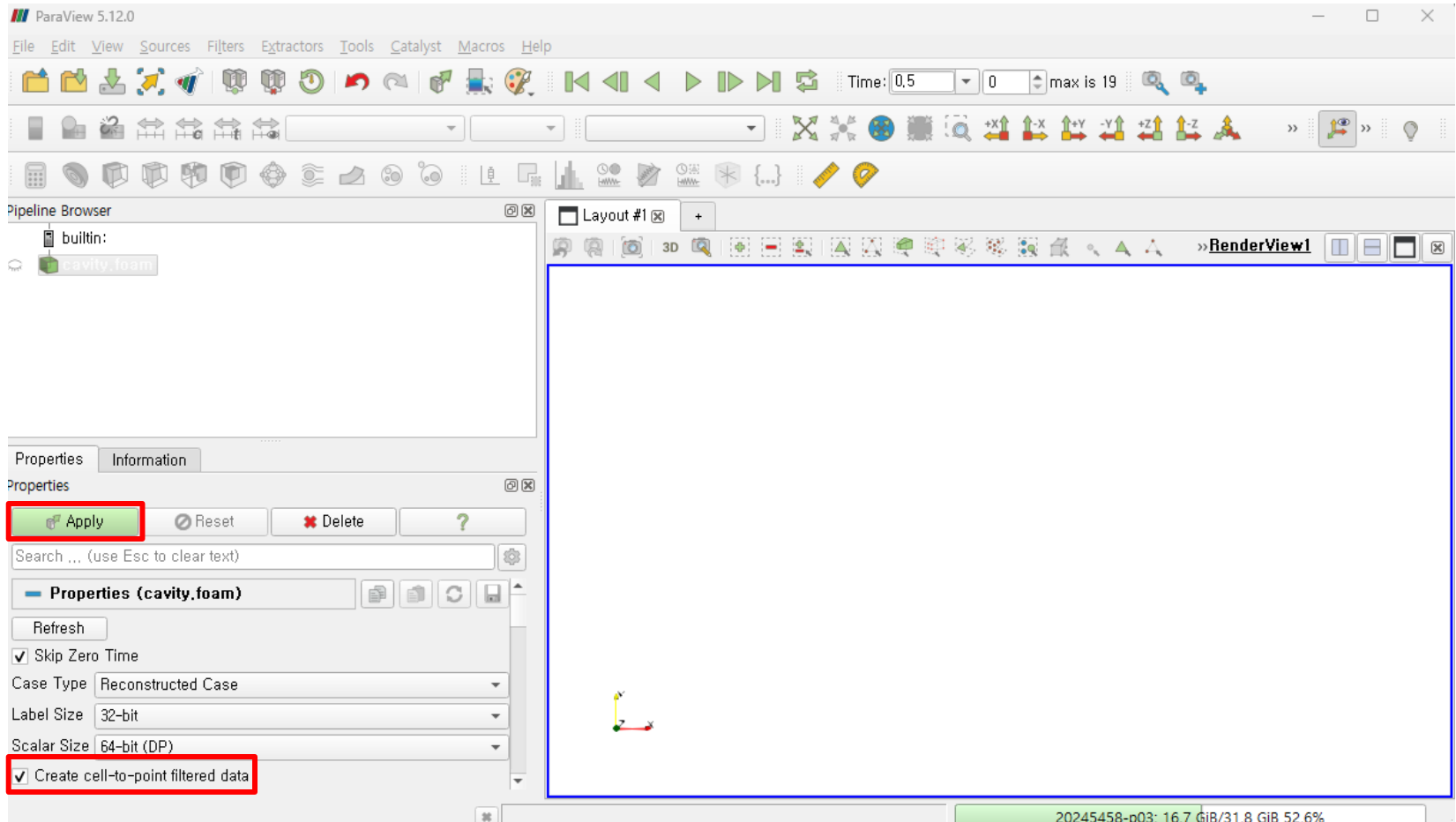
- Choose dummy file (.foam file)



Cavity Flow

- **Post-Processing**

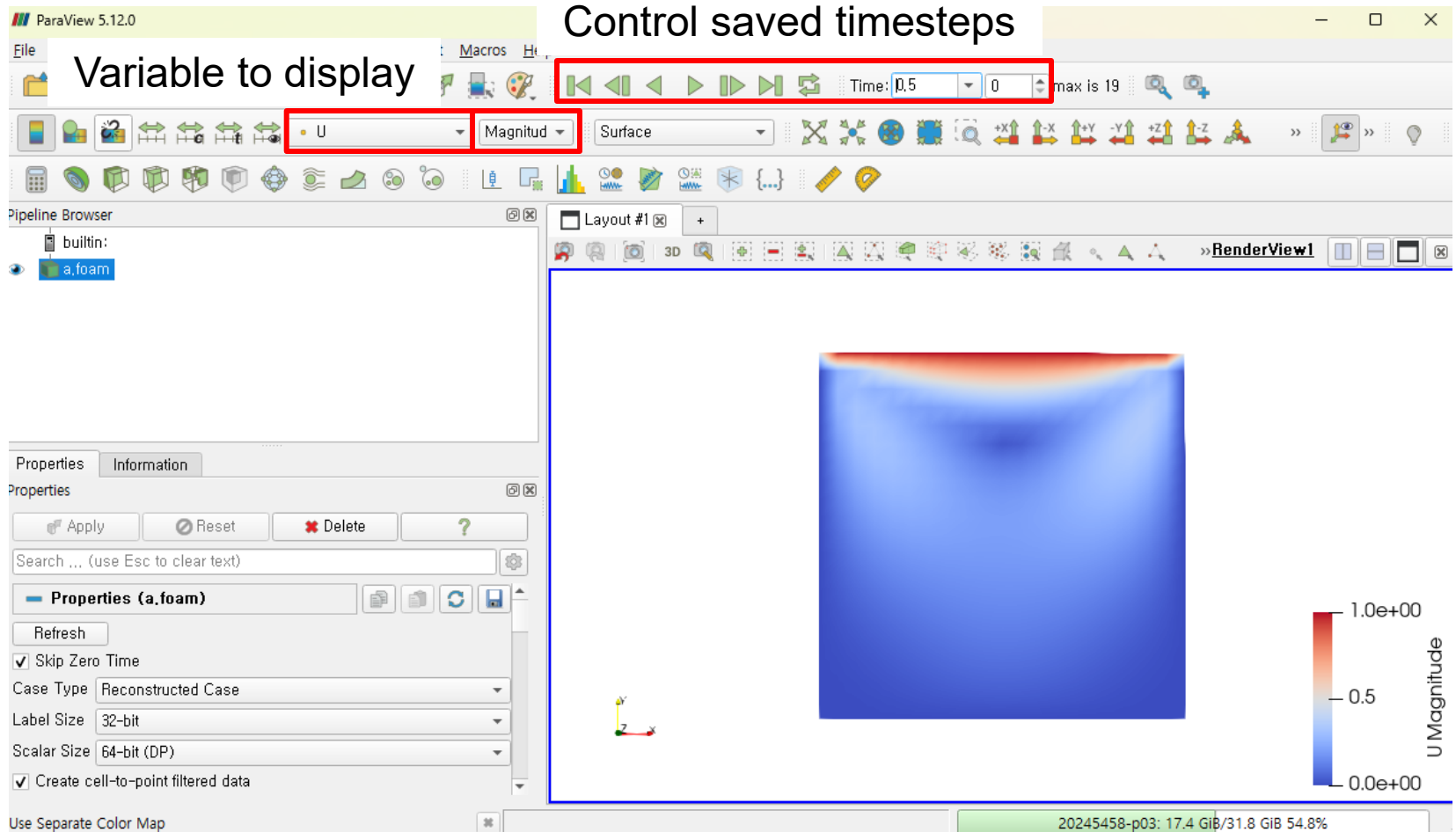
- Press “Apply” to display



Cavity Flow

- **Post-Processing**

- Visualize your simulation



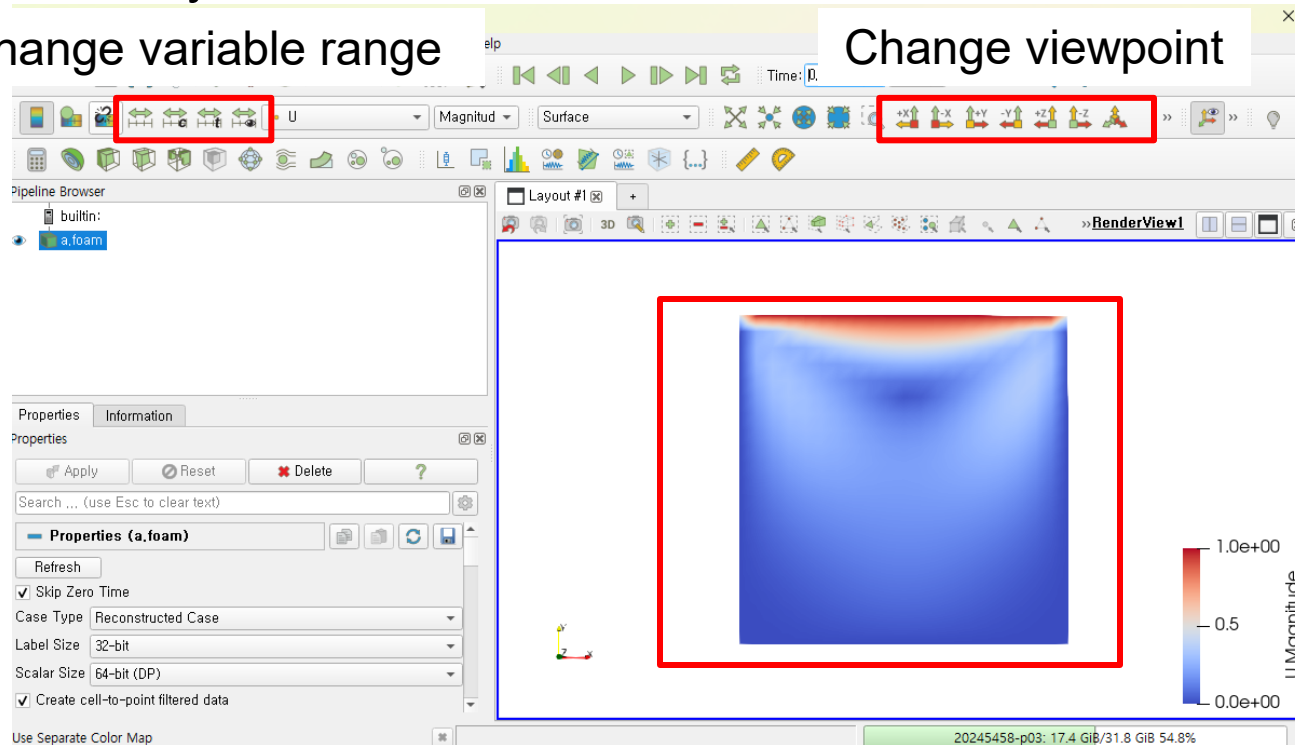
Cavity Flow

- **Post-Processing**

- Visualize your simulation

Change variable range

Change viewpoint



Left click (Ctrl + wheel click/Shift + wheel click) : Change viewpoint

Wheel click (Shift + right click) : move object

Right click (wheel up,down/Ctrl + right click) : Zoom in, out

Shift + left click : rotate object